

Master Business Analytics
2026 - 2027

Professional vs dual variant

Core courses Business Analytics (compulsory)		66 EC						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Machine Learning -> NEW TITLE Mathematical Foundations of Machine Learning	XM_0189	6		6				
Applied Stochastic Modeling	X_400392	6	3	3				
Statistical Models	X_400418	6	3	3				
Research Seminar BA	XM_0054	6				3	3	
Master Project BA	X_400459	36						
Compulsory Professional variant (also in Spec Research)								
Project OBP	X_400213	6			6			
Compulsory Dual variant								
Dual Work period	XM_41010	12						

Specialization: Computational Intelligence		18 EC						
Compulsory courses	Code	EC	P1	P2	P3	P4	P5	P6
Evolutionary Computing	X_400111	6	6					
Deep Learning	XM_0083	6		6				
	XM_0121							
Natural Language Processing Technology		6					6	
Total			18	12	6	3	9	0

Compulsory selection		18 EC (12 EC for dual or professional with semester abroad)						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Data Mining Techniques	X_400108	6					6	
Dynamic Programming and Reinforcement Learning	XM_0093	6		6				
NEW Large Language Models from scratch*	XM_0188	6				6		
Learning Machines**	XM_0061	6			6			6
ML for the quantified self	XM_40012	6						6
Multi-agent Systems	XM_0052	6		6				
Performance of Networked Systems	X_405105	6		6				
Web Data Processing Systems discontinued by MSc CS	XM_40020	6		6				
* machine learning and reinforcement learning (preferably, Advanced Machine Learning and Dynamic Programming and Reinforcement Learning) are necessary prerequisites								
** offered in p3 and p6, limited capacity								

Electives		18 EC						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Asset Pricing*	E_FIN_AAP	6		6				
Advanced Linear Programming (LNMB) not given in 25-26, also not in 26-27	XMM_400326	6				3	3	
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Applied Text Mining 1: Methods	L_PAMATLW004				6			
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Data Mining Techniques	X_400108						6	
Deep Learning	XM_0083	6		6				
Discrete Optimization (LNMB)	XMM_400445		3	3				
Entrepreneurship in Analytics and AI	XM_0090	6				6		
Financial Markets and Institutions** E_FIN_FMI -> renamed into Advanced Financial Markets and Institutions	new code	6		6				
Investments E_EBE3_INVES LEVEL 300 -> NEW COPY of level 400 'Investments for Business Analytics'	XM_0187	6					6	
NEW Large Language Models from scratch***	XM_0188					6		
Learning Machines****	XM_0061	6			6			6
Mathematical Optimization	XM_0051	6		6				
ML for the quantified self	XM_40012	6						6
Natural Language Processing	XM_0121	6					6	
Optimization of Business Processes	X_400422	6				3	3	
Performance of Networked Systems	X_405105			6				
Programming Large-Scale Parallel Systems	XM_40017	6	6					
Project OBP	X_400213	6						
Project Reinforcement Learning	XM_0120				6			
Quantitative Investing*	E_FIN_QI	6	6					
Queueing Theory	XMM_400397					3	3	
Scheduling (LNMB) XMM_400396 -> new name								
Scheduling Theory and Algorithms, from 26-27 on will be offered every other year	XMM_0083	6				3	3	
Stochastic Processes for Finance	X_400352	6	3	3				
Supply Chain Lab	E_TSCM_SCL	6				6		
Time Series Models	E_EORM_TSM	6				6		
*out of Advanced Asset Pricing and Quantitative Investing, only one is allowed								
**course Investments is a compulsory prerequisite								
*** machine learning and reinforcement learning (preferably, Advanced Machine Learning and Dynamic Programming and Reinforcement Learning) are necessary prerequisites								
****offered in p3 and p6, limited capacity								

Specialization: Optimization of Business Processes		18 EC						
Compulsory courses	Code	EC	P1	P2	P3	P4	P5	P6
DPRL	XM_0093	6		6				
Mathematical Optimization	XM_0051	6		6				
Optimization of Business Processes	X_400422	6				3	3	
			0	12	0	3	3	0

Compulsory selection		18 EC (12 EC for dual or professional with semester abroad)						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Linear Programming (LNMB) not given in 25-26, also not in 26-27	XMM_400326	6				3	3	
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Data Mining Techniques	X_400108	6					6	
Discrete Opt (LNMB)	XMM_400445		3	3				
Entrepreneurship in Analytics and AI		6				6		
Evolutionary Computing	X_400111	6	6					
NEW Large Language Models from scratch	XM_0188					6		
Performance of Networked Systems	X_405105	6		6				
Scheduling (LNMB) XMM_400396 -> new name								
Scheduling Theory and Algorithms, from 26-27 on will be offered every other year	XMM_0083	6				3	3	
Supply Chain Lab		6				6		

Specialization: Financial Risk Management		18 EC						
Compulsory courses	Code	EC	P1	P2	P3	P4	P5	P6
Stochastic Processes for Finance	X_400352	6	3	3				
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Investments E_EBE3_INVES LEVEL 300 -> NEW COPY of level 400 'Investments for Business Analytics'	XM_0187	6					6	
			3	3	0	3	9	0

Compulsory selection		18 EC (12 EC for dual or professional with semester abroad)						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Asset Pricing*	E_FIN_AAP	6		6				
Data Mining Techniques	X_400213	6					6	
Derivatives** NEW PERIOD p4->p5	E_FIN_DER	6				6	6	
DPRL	XM_0093	6		6				
Financial Markets and Institutions** E_FIN_FMI -> renamed into Advanced Financial Markets and Institutions	new code	6		6				
Mathematical Optimization	XM_0051	6		6				
Quantitative Financial Risk Management**	E_FIN_QFRM	6					6	
Quantitative Investing*	E_FIN_QI	6	6					
*out of Advanced Asset Pricing and Quantitative Investing, only one is allowed								
**course Investments is a compulsory prerequisite								

Recommendations on the order of courses: Any further edits for 26-27?

- If you wish to follow *Mathematical Optimization* and *Supply Chain Lab*, it is optimal to do *Supply Chain Lab* first and *Mathematical Optimization* next.

- If you wish to follow *Mathematical Optimization* and *Discrete/Continuous Optimization*, it is optimal to do *Mathematical Optimization* first and *Discrete/Continuous Optimization* next.

- If you wish to follow *Project Optimization of Business Processes* and the course *Optimization of Business Processes*, it is optimal to follow the project first and the course next.

- If you wish to follow *Project Reinforcement Learning*, it is optimal to first do *Dynamic Programming and Reinforcement Learning*.

- NEW in 26-27: If you wish to follow *Large Language Models from scratch*, it is optimal to first do *Advanced Machine Learning* and *Dynamic Programming and Reinforcement Learning* (and prerequisites?)

-ADD? Investemtns prerequisites for ...

- It is optimal to follow Deep Learning in year 2.